Project Report: KitaKitar - Snap, Sort, Recycle

1.0 Introduction

1.1 Background of Case Study: Malaysia currently faces a severe waste management crisis, generating over 38,000 tonnes of waste daily, with only 33% of it actually being recycled. A significant barrier to increasing this recycling is called the "Intention-Action Gap". Many people want to recycle, but they just do not know what items should go where because of how many different systems exist as well as local recycling rules variation. This confusion leads to "wish-cycling", which means that up to 25% of items placed in recycling bins are non-recyclable contaminants. They can ruin an entire load of good recyclables. There are many surveys indicating that greater than 60% of people are still confused about what can be recycled and what cannot be recycled.

Chosen SDG: United Nations Sustainable Development Goal (SDG) **13: Climate Action**. This project is trying to reduce landfill waste and support recycling actions.

Problem Statement: Currently, there is a large gap between the public's desire to recycle and their ability to consistently recycle correctly. Without providing accessible educational resources, contamination rates will remain high due to lack of initial incentive to create long-term habits of sustainability.

Objectives: To solve this, KitaKitar serves as a "pocket recycling companion" utilizing a "3C" approach:

- **Clarify:** Use AI to instantly identify waste and remove guesswork.
- **Connect:** Use map integrations to guide users to the nearest appropriate drop-off points.
- **Compensate:** Gamify the experience by allowing users to earn rewards for doing the right things.

2. Review of Existing Solutions

While there are existing waste management apps and municipal websites, most current solutions suffer from the following gaps:

1. Static Information: Many solutions are relying on manual searches or reading long lists to find specific types of trash (therefore not providing enough immediate clarity when you are holding a piece of trash in your hand),
2. Lack of Incentive: The majority of existing solutions provide little to no incentive to develop sustainable recycling habits;
3. No Dynamic Assistance: Most of them do not provide a feature that can answer complex questions of recycling (e.g., "How do I recycle a pizza box with grease on it?").

A solution like KitaKitar will fill these gaps through the combination of on-the-spot AI visual identification, a chatbot-based interactive experience and a gaming incentive/rewards program.

3. Methodology

Technical Track

KitaKitar is built using a **client–cloud architecture** that supports real-time interaction, scalability, and cross-platform mobile deployment.

1.Frontend – Flutter

Enabling a single codebase for Android and iOS. Flutter handles:

- User authentication flows
- Camera-based waste scanning
- Map visualization
- Leaderboards, profiles, and QR scanning
- State management using Provider

2.Backend – Firebase

- **Firebase Authentication**
  Supports Email/Password and Google Sign-In for secure user access.
- **Cloud Firestore**
  Stores user data, recycling centers, materials, scan results, points, and leaderboard data.
- **Firebase Storage**
  Stores uploaded images related to waste scanning.
- **Cloud Functions**
  Handles server-side logic such as point calculation, QR validation, and leaderboard updates.
- **Firebase Analytics**
  Tracks feature usage and user interaction for impact analysis.

3.AI Layer – Google AI Technologies

The application is designed to integrate **Google AI services** (Gemini AI) for photo-based waste recognition and AI Chatbot.
Currently, for Photo Recognition, we use Gemini 2.5 Flash due to its price and accuracy; for AI Chatbot we use Gemma 3 27B because of its speed and huge limit rates.

4.Location Services – Google Maps

Google Maps SDK is integrated to:

- Display nearby recycling centers
- Filter centers by accepted materials
- Provide location-based user interaction

Implementation Details

- Users authenticate using Firebase Authentication.
- The scan feature uses the device camera to capture waste images.
- Images are uploaded to Firebase Storage.
- AI scan results are stored in Firestore under /ai_scans.
- Recycling centers and accepted materials are queried from Firestore and displayed using Google Maps.
- QR codes are scanned to validate recycling actions and reward points.
- Leaderboards aggregate data from /transactions and /leaderboards collections.
- Cloud Functions ensure secure point allocation and prevent QR reuse.
- The app is structured for maintainability with clear separation between UI, services, and data models.

Challenges Faced

Challenge 1: AI Integration Readiness

Integrating real-time AI recognition requires careful handling of latency, image size, and API costs.

Solution:
We implemented a mock AI response layer while preparing service abstraction that allows seamless future integration with Gemini AI.

Challenge 2: Secure Reward Distribution

QR-based reward systems are vulnerable to duplication and abuse.

Solution:
We implemented one-time QR codes stored in Firestore and validated via Cloud Functions before awarding points.

Challenge 3: Managing Complex Firestore Data

Handling multiple collections (users, centers, materials, transactions) required careful data modeling.

Solution:
We designed a structured Firestore schema with clear relationships and role separation to support scalability.

4. Results

KitaKitar achieved its core objective of making recycling easier and creating a rewarding recycling experience.

User Feedback:

Initial testing got a lot of positive feedback about using KitaKitar. For example, some users found that the app makes recycling much easier and gives them confidence in how to sort items for recycling. At the same time, the gamified reward system also motivated users to recycle more consistently.

Social Impact:

- Democratizing Green Knowledge: KitaKitar's ability to allow people to recycle by simply taking a picture helps provide equal access to recycling information for people.
- Building Sustainable Habits: By gamifying the experience of recycling, KitaKitar has been able to change people's perceptions about recycling from being a chore to an enjoyable part of their daily routine.
- Cleaner Communities: KitaKitar's ability to reduce the amount of waste going to landfills and the amount of "wish-cycling", create cleaner communities from the beginning of recycling.

5. Prototype Showcase

5.1 User Sign-In

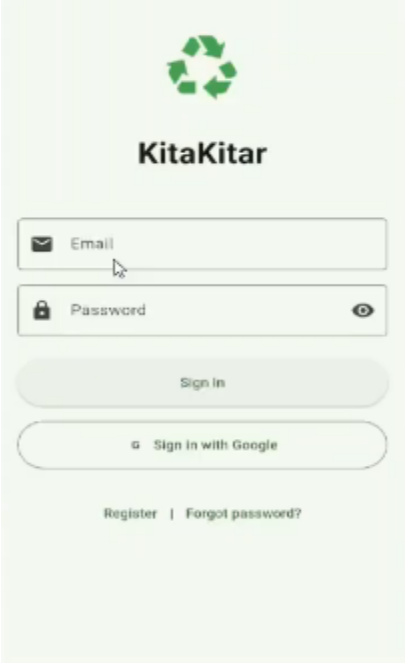

Image 5.1.1: Mobile application user sign-in page.

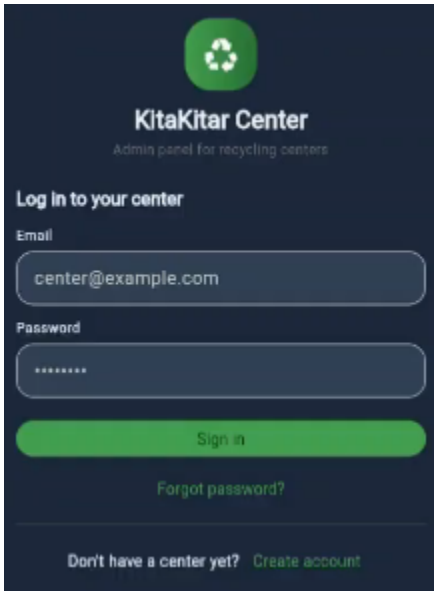

Image 5.1.2: Website user sign-in page.

First of all, the general public will be using the mobile application which we can see at Image 5.1.1. They will be able to use the legacy method of email and password, or use the “Sign in with Google” option. For recycling centers, they will be signing in on our website page that is dedicated to them. For first timers, they will have to register first through our “Register” option and fill in required details as prompted.

5.2 User Mobile Application Walkthrough

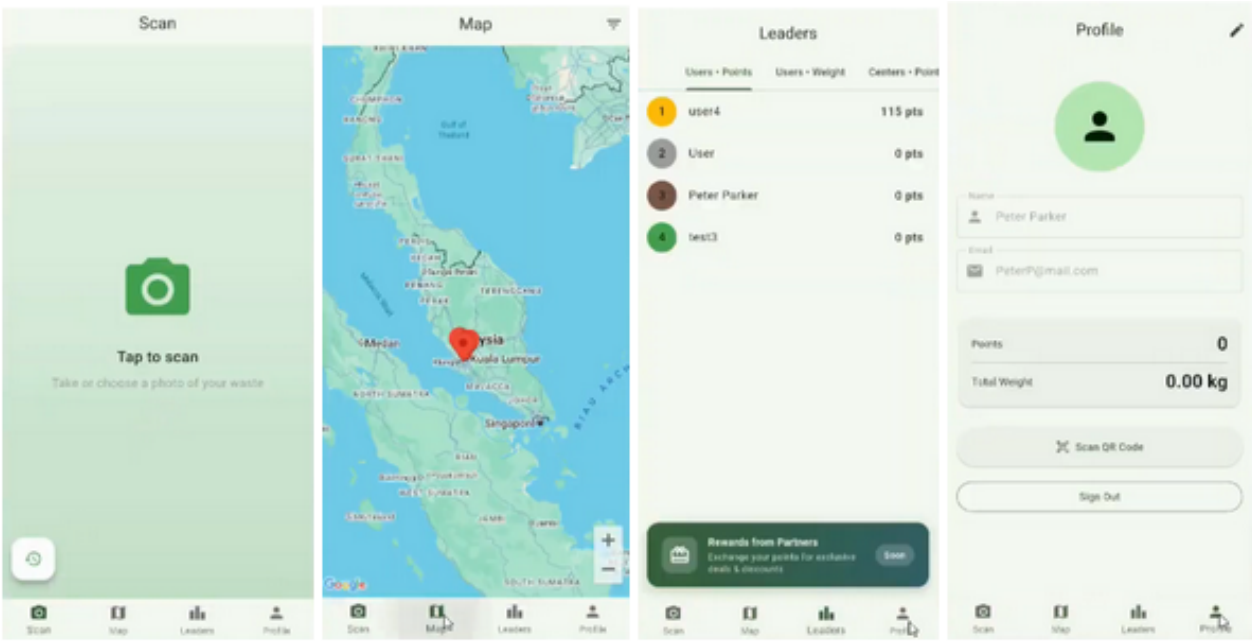

Image 5.2.1: The 4 Main Pillars (Scan, Map, Leaders, Profile).

Users will be able to access four functions as shown, scan, map, leaderboard and profile.

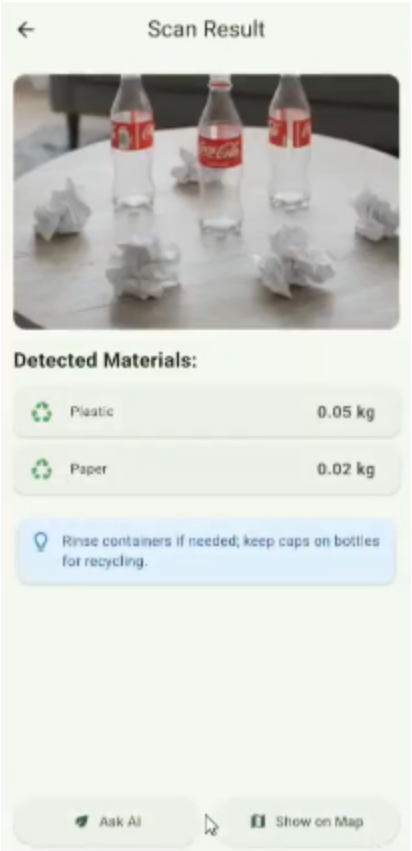

Image 5.2.2: Scanned Results.

Once users scanned their recyclables, our AI will process the image and produce a result. This action can be further used to ask our AI chatbot regarding details about their recyclables and more, or users can click “Show on Map” to find the nearest recycling center that accepts the material of scanned recyclables.

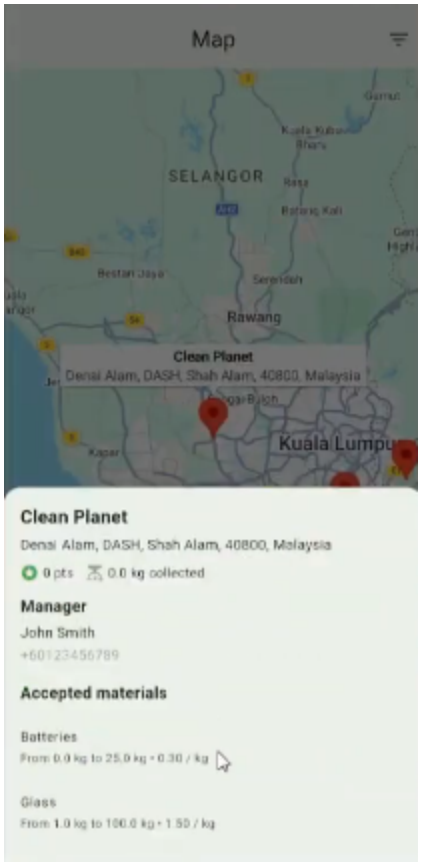

Image 5.2.3: A specific recycling center with detailed information.

Once they choose a desired recycling center, they can head over there and bring their recyclables to the recycling center. The manager of the recycling center will generate a QR code for them to scan and obtain points for recycling.

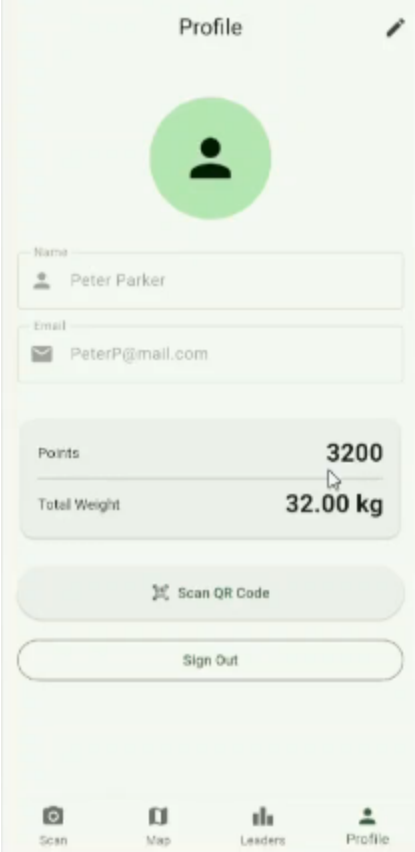

Image 5.2.4: Peter Parker obtained 3200 points after scanning a dedicated generated QR code.

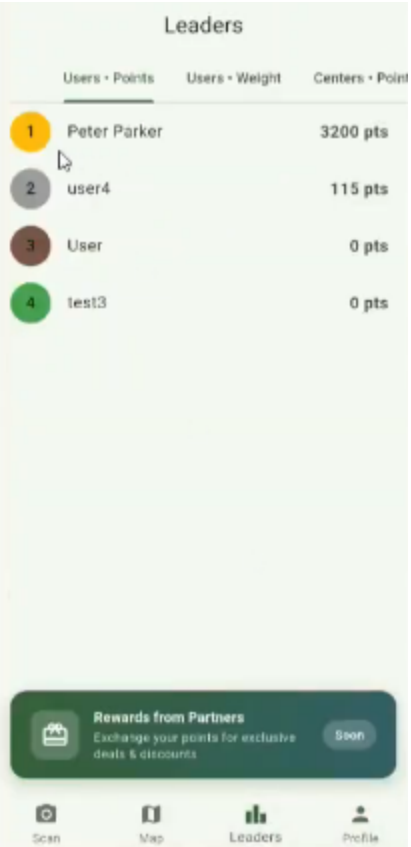

Image 5.2.5: Points obtained/ Weight of recycled items will also be reflected in the leaderboard.

After users scanned the generated QR code of their recyclables, the points obtained will be updated in the leaderboard as well. Which in the future, can be used to redeem rewards from various partners. One thing to notice is that this leaderboard also provides a competition for weight of recyclables recycled as well, for both users and recycling centers. However, the recycling center will be awarded through points given to customers.

5.3 Recycling Center Website Walkthrough

Image 5.3.1: Central Dashboard of a specific recycling center.

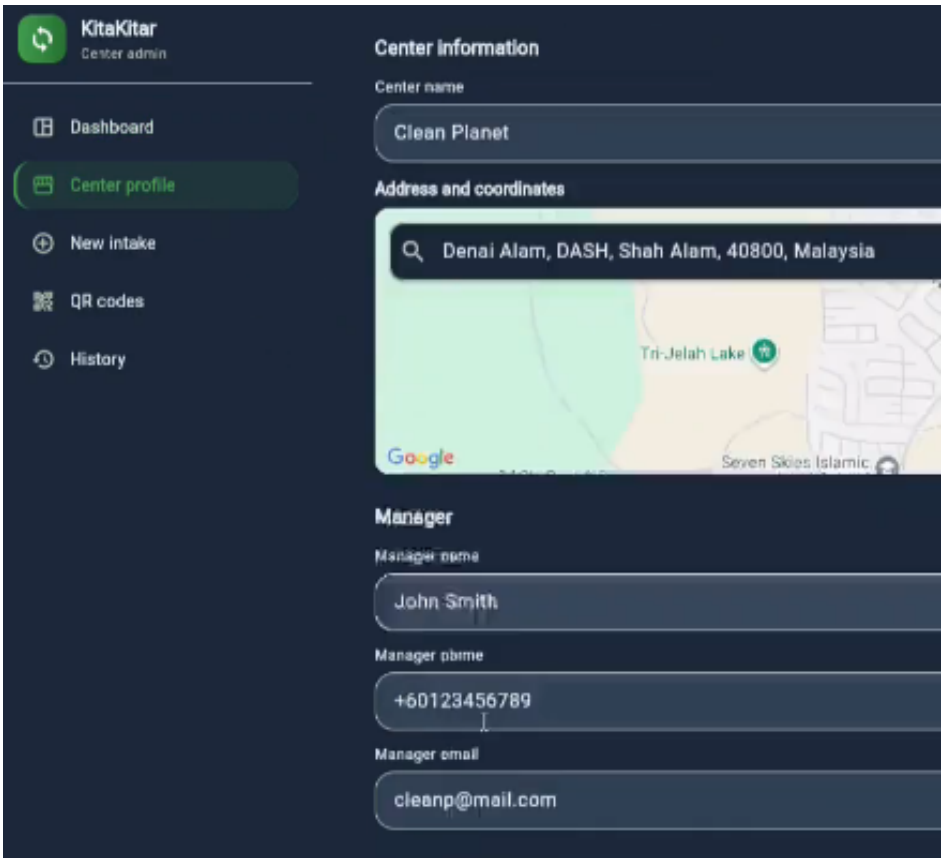

Image 3.5.2: Center Profile Details.

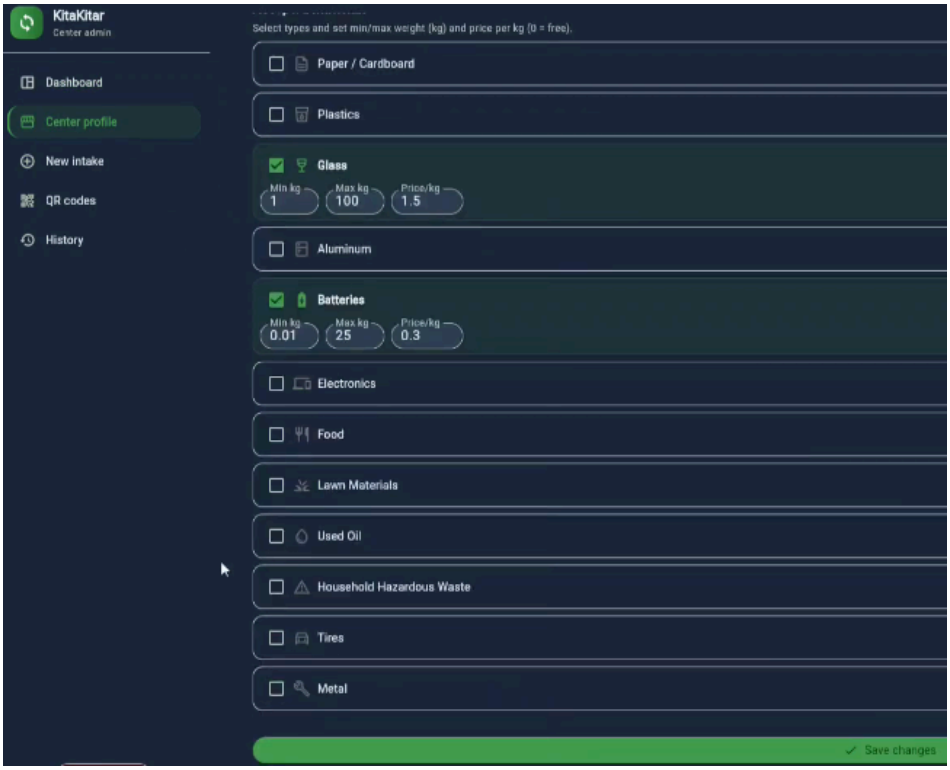

Image 3.5.3: Accepted Materials under Center Profile Details.

Image 3.5.4: New Intake Selection.

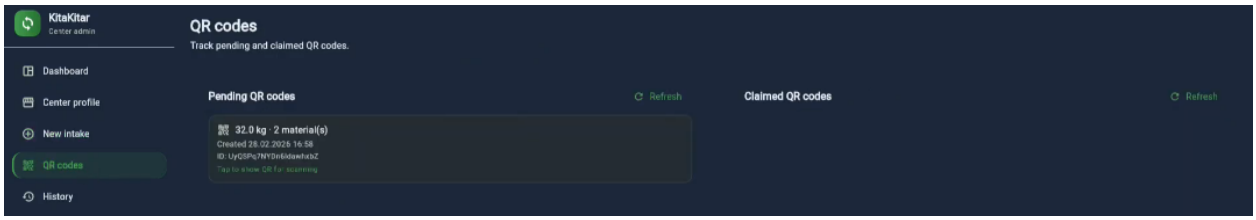

Image 3.5.5: Pending and Claimed QR Codes.

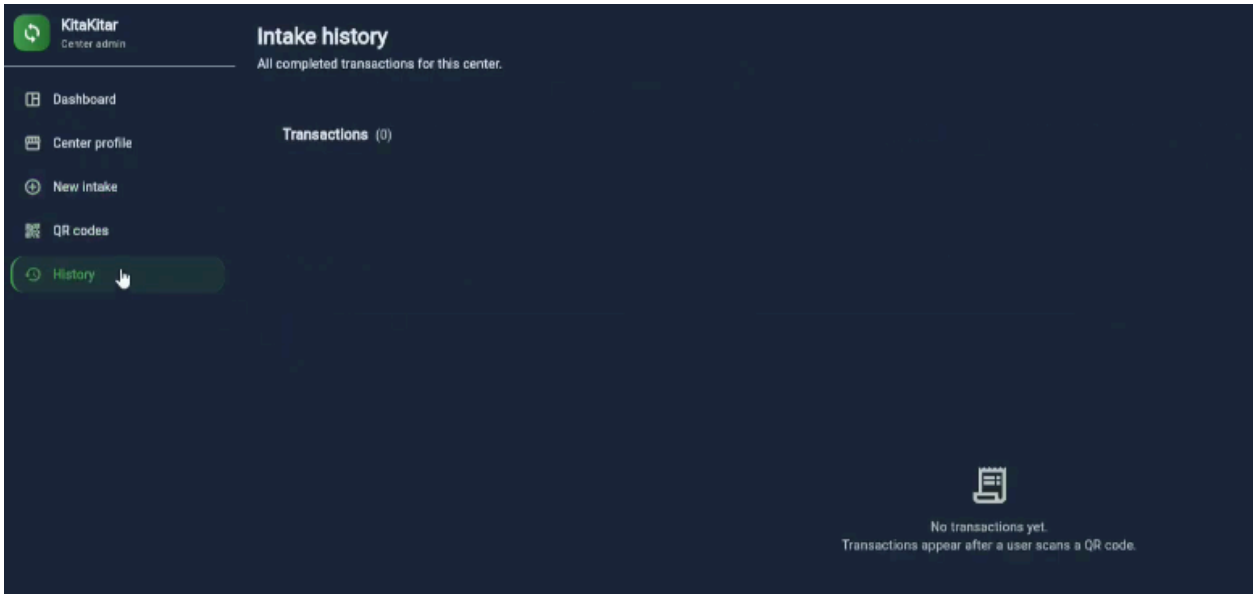

Image 3.5.6: Intake History Tab.

Once the recycling center manager registered and signed in, they will first stumble across our main dashboard as shown in Image 5.3.1. It has a few core information listed out and it also provides a trend chart of intakes in the past 14 days. In our center profile section, recycling centers will be able to change their details of their center name, location, manager details and such as shown in Image 3.5.2. Underneath it, recycling centers can add/remove accepted materials and update the rates based on their needs and regulations like in Image 3.5.3. When someone comes and recycles, they can open the “New Intake” tab and enter the weight of the recyclables, then click “Generate QR code” to let the customer scan in order to obtain points. After that, every QR code will be placed into the “QR codes” tab, they will either be in the “Pending” section or “Claimed” section depending on the status of the QR code. Lastly, every intake/transaction will be logged into the “History” tab after a user scans a QR code.

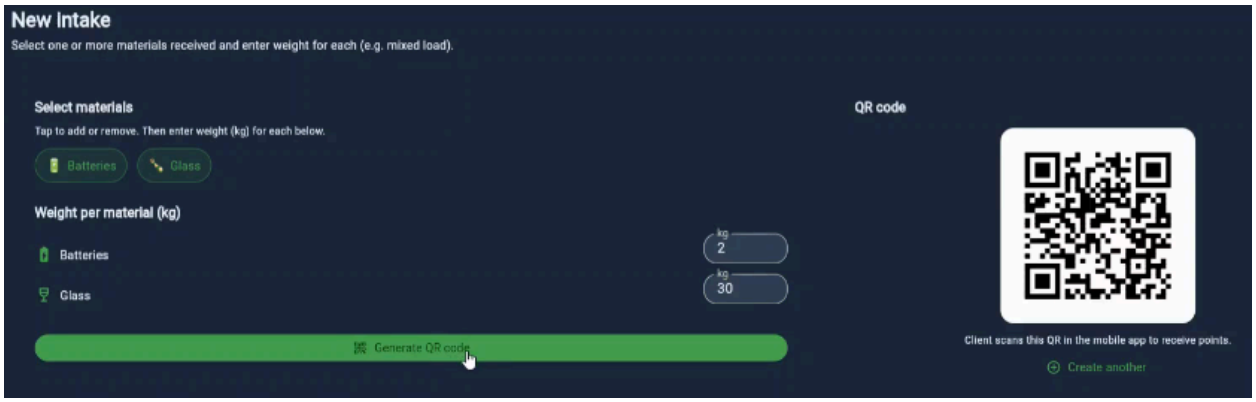

Image 3.5.7: Example of generating a QR code for a customer.

6. AI Acknowledgement

- **Gemini / ChatGPT:** Utilized for brainstorming application flows, structuring the final project report and generating debugging logic for Flutter.
- **GitHub Copilot:** Utilized within the IDE for real-time code auto-completion.
- **Gemini 2.5 Flash & Gemma 3 27B:** Core infrastructural APIs utilized directly within the application's runtime for image classification and chatbot features.

7. Conclusion & Future Roadmap

KitaKitar demonstrates how using an accessible AI platform along with an instant reward system can successfully overcome the intention to act towards proper waste management. We can significantly decrease contamination and develop a sustainable circular economy by helping users at the time they are disposing of their waste.

Future Roadmap:

- **1st phase AI Improvement:** Enhance waste recognition accuracy using local Malaysian waste data and packaging.
- **2nd phase Smart Physical Scanner:** Deploy Arduino/ESP32-based recycling scanners at recycling centers to verify disposal and auto-reward users via QR or NFC.
- **3rd phase Scalability:** Extend the current Firebase-based architecture to support city-wide deployment and smart city integration with minimal changes.